# Who Chooses Your Hospital? Referral Capital, Physician Agency, and the Measurement of Hospital Competition

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## The object that replaced the mechanism

**Referral capital**: a relationship-specific asset, owned by the referring *organisation*, that determines hospital demand. Not an incentive — an asset.

# Why This Matters: Merger Review Runs on a Demand Model

**The standard pipeline (Capps–Dranove–Satterthwaite; Gowrisankaran et al.):** estimate patient hospital choice on distance, quality and price → compute willingness-to-pay and *diversion ratios* from those preferences → diversion decides whether a merger is challenged.

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**Every step assumes the patient is the decision-maker; in managed care she is not. No one chooses the hospital at any single moment.** She never compares hospitals; the PCP picks a *specialist*, not a building; the surgeon uses his usual site. A patient-only model cannot see that, so it loads the routing onto her taste for distance.

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**How much does the referring organisation move the destination?** No model needed:

| Her next operation is at a *different* hospital when. . . | raw | matched on elapsed time to that operation |
|---|---|---|
| she keeps her practice | 62.5% | 55.3% |
| she changes to a practice that **does** use her old hospital | 64.1% | 59.1% |
| she changes to a practice that has sent **nobody** there | **84.1%** | **77.5%** |

⇒ changing practice moves her **4 points**; changing to one with a *different hospital list* moves her **19 more**. [23]

# The Same Evidence Supports Two Papers — I Would Value Your View

**(A) A MEASUREMENT paper**

*The instrument is miscalibrated.* Demand models attribute physician routing to patient preference; correcting it moves diversion and WTP.

+ closest to what I can defend line by line
+ the nulls on money make it the honest frame
− the referee asks "so what" and gets a correction factor

**(B) An IO paper**

*Here is an asset, here is who owns it, here is what its ownership does to competition.* Market power in hospital markets is partly **control of referral chains**.

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## Where I currently land

Keep measurement as the **method**; make market structure the **claim**. **Hospital demand is mediated by a relationship-specific asset owned by physician organisations. Merger analysis measures demand as if patients chose, so it misplaces where market power sits — and because disruption reallocates that asset toward systems, concentration rises with no merger at all.**

The next four slides are the evidence for that sentence; slides 12–14 are the three IO results it rests on.

# Data: California All-Payer Claims, 2018–2023

**HCAI Health Payments Database**, commercial insurance, ages 18–64.

| | |
|---|---|
| Members in the universe | 8,471,732 |
| Claim headers / service lines | 184.3M / 244.1M |
| Doctor visits (encounters) | 187.6M |
| Elective facility operations | 5,068,724 |
| **Patients who change PCP practice, operation both sides** | **114,937** |

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**One design decision makes the paper possible: the data is extracted for people, not for operations.**

- An earlier build pulled records only for patients who had surgery — so a person exists in the data *because of the outcome being studied*.
- That destroys every within-person comparison: **83,037** usable within-member insurance changes collapsed to **582**; the mover design lost 258,000 patients to 4,800.
- Re-extracting member-anchored recovered them. **6.0**× the clean movers.

# Fact 1: Patients Do Not Go to the Nearest Hospital, and Practices Disagree

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Total variation distance between two organisations' hospital distributions, same procedure group:

| | mean TVD |
|---|---|
| *Same* organisation, split in half (pure sampling noise) | 0.356 |
| Different organisations, **same ZIP3** | **0.778** |
| Different organisations, different ZIP3 | 0.988 |

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## Why the first row is the whole trick

A raw TVD of 0.78 is uninterpretable — two *identical* practices differ by chance, and by more when they are small. So I measure the null in the same data: split one practice in half. **Putting two practices on the same streets removes only a third of the gap.**

[20]

# Fact 2: It Is Not Geography — the Twin-Organisation Benchmark

**The objection.** The menu is built from an organisation's *other* patients, and those patients live near each other. So it could just be encoding "hospitals used by people around here."

**The design.** For each episode, recompute the menu from a **different organisation in the same ZIP** — same geography, same case mix, nothing else shared. Available on 3.74M episodes.

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| predictor | picks the realised hospital | top-3 |
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| a twin organisation, same ZIP and group | **0.130** | 0.269 |
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## The second comparison is the sharper one

Own beats twin 3.5×. But **the twin does not beat pure proximity at all** (0.130 vs 0.126): a rival organisation on the same street adds *nothing* over guessing the nearest hospital. All of the predictive content is in *her* organisation specifically.

[51] — leave-one-out because the raw own share counts her own episode; the twin never can.

# The Design: Patients Who Change Practice

Practice $A$ sends 40% of its patients to hospital $h$; practice $B$ sends 5%. A patient moves from $A$ to $B$. **How much of that 35-point gap travels with her?**

$$\delta = \frac{\text{change in her probability of using } h}{\text{gap between } A \text{ and } B}$$

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**Answer:** $\boldsymbol{\delta \approx 0.28}$ at the specialist organisation, and **0.46–0.50** at the hospital itself. **Every correction for measurement error pushes it UP** — 0.33 where two independent switch definitions agree, 0.35 on decisive switches, 0.30 where the insurer independently confirms the switch. **So 0.28 is a floor, and that is attenuation, not specification search.**

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**The donut width used to be a free parameter. It no longer is.** The baseline is now defined by an *event* — episodes before her first visit to $B$, which is when drift toward $B$ can begin — rather than by a chosen number of months. Jump **0.497**, no window to pick. [60, 22, 40]

# Is It Causal? The Objection Is Sorting

Patients who like hospital $h$ might simply pick practices that use $h$. Four answers, none of which requires me to model selection:

| | jump | |
|---|---|---|
| Voluntary switches | 0.430 | |
| **Forced** — practice closed, doctor left the market | **0.432** | **difference CI spans zero, both clustering levels** |
| Pseudo-movers: never switched, fake date + fake destination | **−0.091** | opposite-signed; no jump at the fake event |
| Internal placebo: practice re-badges at the same address | **+0.027** | CI spans zero |
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**Patients who had no choice transfer exactly like patients who chose.** And the horizon sweep explains the one thing that looked like selection: the forced–voluntary gap runs $+0.071$ at one year, $+0.026$ at two, $-0.002$ **at three**. It was take-up completing, not a different effect.

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**The falsification test that could have killed it:** the menu's pull should weaken where gatekeeping is weak. It does — coefficient **12.29 under HMO against 9.76 under PPO**. [40, 42]

# What Separately Identifies Distance from the Menu

Two things can change for a patient: **where she lives**, and **which practice she uses**. They are independent shocks in this data — physician mobility and residential moves overlap on only **1.19%** of cases.

| P(same hospital as before) | keeps her practice | changes practice |
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## A double dissociation

**A move changes how far her hospital is and not which one it is. A switch does the reverse.** Most demand papers assert this separation; here it is measured — which is what lets a distance parameter and a menu be estimated at the same time.

[26]

# Fact 3: It Is a Consideration Set, Not a Pass-Through

Ask the return question *conditional on whether her new practice uses that hospital at all* (leave-one-out, so her own episodes cannot define the menu):

| Her old hospital is... | she returns to it |
|---|---|
| **off the new practice's menu** — they have sent *nobody*, ever | **0.159** |
| <1% of their volume | 0.177 |
| 1–5% | 0.273 |
| 5–15% | 0.318 |
| **>15% — a core hospital for them** | **0.581** |
| **never switched practice at all** (benchmark) | **0.375** |

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## Read the first, last and benchmark rows together

Switching practice *per se* does little — on-menu switchers sit **3.5–5 points** below people who never switched, in every elapsed-time band. **Switching to an organisation that does not use your hospital does four times as much** — a further **19–21 points**.

And distance is *flat* inside the off-menu band (0.158 / 0.160 / 0.160) — the menu is not a proxy for geography. [23, 25]

# IO Result 1: It Behaves Like a Relationship-Specific Asset

If referral capital is an *asset*, it should **accumulate**, and it should be destroyed by the thing that severs the relationship rather than by personnel turnover.

| how long the new attachment lasted | jump | members |
|---|---|---|
| under 1 year | 0.560 | 55,050 |
| 1–2 years | 0.640 | 26,198 |
| 2–3 years | 0.743 | 11,673 |
| 3+ years | *0.564* | *5,794* |
| **never switched again** | **0.756** | — |

The 3+ cell breaks the monotonicity on 5,794 members and I show it rather than hide it; the censoring test passes and *steepens* the gradient.

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## Why this is Klein–Crawford–Alchian, not just a gradient

A specific asset creates quasi-rents; quasi-rents invite appropriation; **vertical integration is the response.** That is exactly why hospitals buy physician practices — they are buying an input to bargaining leverage. It also predicts the falsification that holds: a *forced* second switch gives **0.488**, **below** voluntary's 0.561 — the opposite of what match-quality would predict.

[28]

# IO Result 2: The Asset's Boundary *Is* the Firm's Boundary

Where exactly does routing change? Three nested comparisons:

| what changes | $\delta$ | difference |
|---|---|---|
| same organisation, **same building** | 0.063 | **0.010 ± 0.014 — indistinguishable** |
| same organisation, **different building** | 0.073 | |
| **different organisation** | 0.159 | **0.090 ± 0.012** |
| practice merely **re-badges** (new billing ID, same address) | | **+0.027, CI spans zero** |

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## Bracketed from both sides

Coarser than the firm (a smoothed "practice" spanning owners) **fails**; finer than the firm (organisation × site) **adds nothing**. **The boundary of the routing asset coincides with the boundary of the firm** — not the building, not the individual physician, not the health system.

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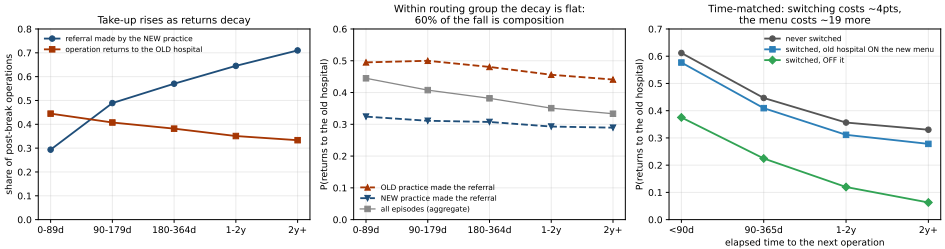
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**Why an IO audience should care.** It says the asset is *owned*, and tells you by whom. A switch that is administratively real and relationally empty moves nothing — which is also the cleanest internal placebo in the paper. [12, 40]

# The Asset Transfers Gradually — and I Can Show Why




**Left:** the new practice's share of referrals rises 0.29 → 0.71 while returns to the old hospital fall 0.44 → 0.33. **Centre:** within a routing group the decay nearly vanishes — **60% of the aggregate fall is composition**, the old practice's referral pipeline emptying, not habit erosion. **Right:** time-matched, switching costs ∼4 points and the menu costs ∼19 more. [41, 23]

# Three Alternative Explanations, Each Killed Against a Measured Null

| Objection | Design | Result |
|---|---|---|
| **"It's the insurance network"** | Same organisation, *different payers* — identity fixed | across payers TVD **0.414** vs noise floor 0.291; *between* organisations 0.739. The menu travels with the firm, not the contract |
| **"It's geography"** | A *twin* organisation in the same ZIP | own org **0.458**, twin **0.130** — and the twin does not beat *guessing the nearest hospital* (0.126) |
| **"It's money"** | Capitation and patient price exposure, within organisation | **0.299 vs 0.298**. A clean null — and the same machinery finds large effects elsewhere, so it is a measured null, not a failure to measure |

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## A habit I would like to defend in the paper

Every comparison is reported against a null **measured in the same data, with the same estimator, at the same cell sizes**. It converts "is 0.41 big?" — an argument about priors — into "is 0.41 bigger than the 0.29 identical objects produce here?" The floors also pass their own test: they *fall* with cell size as sampling theory requires, while the signal stays flat.

[50, 51, 52]

# The Payoff: A Diversion Ratio Measured Rather Than Simulated

Merger review needs *diversion* — if a hospital were removed, where does its demand go? Normally simulated from an estimated demand model. **Here it can be measured: when a referring practice dies, its patients must be re-routed by somebody.**

| | share of departing demand | vs. availability |
|---|---|---|
| Stays inside the origin hospital's **system** | **0.392** | **1.5–3.0×** |

Benchmarked on the same-system share of *the patient's own choice set*: a system owning 60% of nearby hospitals would capture 60% by chance. **The ratio is radius-sensitive and I report the range** — 1.46× on the 5 nearest, 2.96× on the 50-nearest default. The numerator cannot move, only the benchmark. **It exceeds 1 under every rule**, which is the robust claim; the magnitude is not.

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## Why this matters even if the demand model never gets built

This is the quantity merger review wants, obtained from an **exogenous supply shock** with no preferences estimated and no model imposed. If the structural stage slips, the paper still has a diversion result.

[63]

# IO Result 3: A Consolidation Ratchet, Measured Rather Than Assumed

Split the same shock by **who owns the origin hospital**. Excess retention over what availability alone would give:

| origin hospital is... | excess retention | |
|---|---|---|
| **in a health system** | $+0.157$ | **[+0.152, +0.162] — excludes zero** |
| **independent** | $-0.005$ | **[−0.017, +0.006] — spans zero** |

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| **independent** | $-0.005$ | **[−0.017, +0.006] — spans zero** |

## The claim this licenses

Dislodge a patient from a *system* hospital and the system pulls her back well above chance. Dislodge her from an *independent* and she is re-allocated at *exactly* the rate availability predicts — no better, no worse.

**Every disruption is an opportunity for a system to gain and none for an independent.**

# IO Result 3: A Consolidation Ratchet, Measured Rather Than Assumed

Split the same shock by **who owns the origin hospital**. Excess retention over what availability alone would give:

| origin hospital is... | excess retention | |
|---|---|---|
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**Iterate that over practice closures, retirements and physician exits and you get concentration with no merger at all** — through a channel merger review cannot observe, because there is no transaction to review. That is the most policy-relevant thing in the project, and it needs no structural model. [63]

# What I Do *Not* Have — Stated Before You Ask

- **The employment channel is suggestive, not established.** Acquired practices' routing following the acquirer (cf. Cooper et al. 2025) shows a DiD of $+0.051/+0.073$ against a **pre-trend of** $-0.089$. A DiD the size of the pre-trend is a trend. *This is the result an IO framing would put most weight on, and it is the weakest one I have.*

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**Why I am telling you this on a slide:** the same audit that produced the results above found four defects in my own code this month, each invisible in a single run. The discipline is the contribution as much as the numbers.

# Where It Goes — and Three Things I Would Like Your View On

**Built and audited:** the member-anchored panel, the mover design, the menu, the falsification battery, 239M-row choice sets with the referral menu and price exposure joined on. All 30 analysis scripts re-run and current.

**Next six months:**

1. **M0 → M1 → M2.** Patient-only logit; menu-in-utility; then a two-stage consideration-set model where the chain generates the menu and patient utility chooses within it.
2. **Re-derive diversion and WTP under M2**, and re-run a merger screen to ask how many verdicts flip.
3. **(If the IO framing wins)** link referral concentration to *negotiated prices* — 379,078 payer × facility cells are built. That is the step that turns "diversion is mismeasured" into "market power is partly control of referral chains".

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**Three questions:**

- **Framing.** Measurement or IO (slide 4)? I lean IO-claim / measurement-method, but it raises the bar on the acquisition result.
- **How hard to push the policy payoff.** The merger-screen counterfactual is what a referee wants and is the part not yet built. The measured diversion on slide 16 is a fallback needing no model.
- **The variation I am not using.** Exits and closures are clean but small (∼5,600 movers). Is forced-vs-voluntary equality enough, or should the paper lean harder on the forced sample?